

## Resettable Fuse PTC 30V Series

### Features

RoHS Compliant & Halogen Free

Radial leaded Devices

Cured, flame retardant epoxy polymer insulating material meets UL94V-0 requirements

Operation Current: 0.5A~9A , Maximum Voltage: 30Vdc, Operating

Operating Temperature: -40℃ to +85℃

Agency recognition:



### Dimensions(Unit:mm)

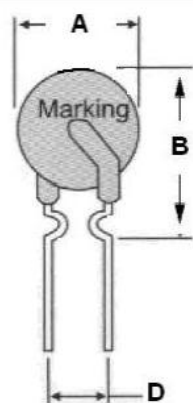


Fig.1

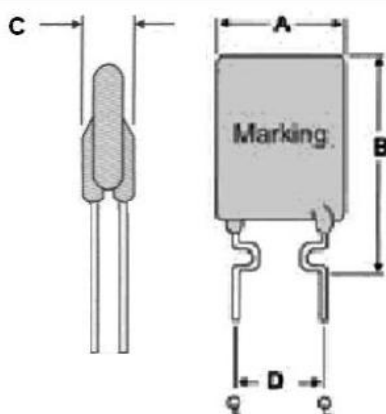


Fig.2

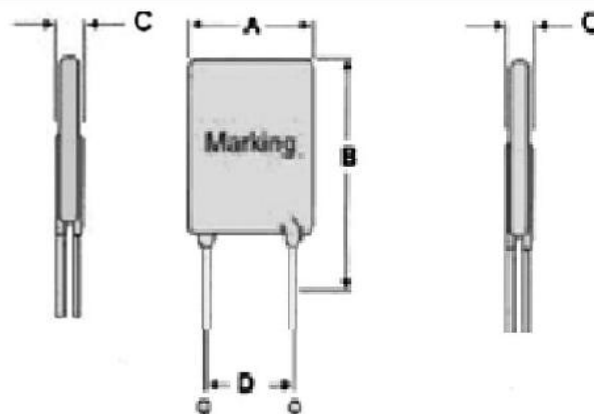


Fig.3

| Part number | Dimensions(mm) |        |        |        | Lead material    | Shape | Certification |     | Delivery Time |         |
|-------------|----------------|--------|--------|--------|------------------|-------|---------------|-----|---------------|---------|
|             | A(max)         | B(max) | C(max) | D(Typ) | Tinned Matel(mm) | Fig   | UL            | TUV | in stock      | Produce |
| JK30-050    | 7.40           | 12.7   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig1  | -             | √   | 3days         | 14days  |
| JK30-075    | 7.40           | 13.0   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig1  | √             | √   | 3days         | 14days  |
| JK30-090    | 7.40           | 18.5   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | √             | √   | 3days         | 14days  |
| JK30-110    | 7.40           | 18.5   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | √             | √   | 3days         | 14days  |
| JK30-120    | 7.40           | 18.5   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | -             | √   | 3days         | 14days  |
| JK30-135    | 9.20           | 17.6   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | √             | √   | 3days         | 14days  |
| JK30-160    | 9.20           | 20.2   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | √             | √   | 3days         | 14days  |
| JK30-185    | 9.20           | 20.2   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | √             | √   | 3days         | 14days  |
| JK30-200    | 15.2           | 20.2   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | √             | √   | 3days         | 14days  |
| JK30-250    | 13.2           | 22.4   | 3      | 5.10   | 24 AWG/ Φ0.5     | Fig2  | √             | √   | 3days         | 14days  |
| JK30-300    | 13.2           | 20.4   | 3      | 5.10   | 20 AWG/ Φ0.8     | Fig3  | √             | √   | 3days         | 14days  |

Specifications are subject to change without notice

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Tel: +86-755-27465585

HUAAN LIMITED

[www.huaandz.com](http://www.huaandz.com)

Email: [sales@huaandz.com](mailto:sales@huaandz.com)

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| Part number | Dimensions(mm) |        |        |        | Lead material    | Shape | Certification |     | Delivery Time |         |
|-------------|----------------|--------|--------|--------|------------------|-------|---------------|-----|---------------|---------|
|             | A(max)         | B(max) | C(max) | D(Typ) | Tinned Matel(mm) | Fig   | UL            | TUV | in stock      | Produce |
| JK30-400    | 14.0           | 23.7   | 3      | 5.10   | 20 AWG/ Φ0.8     | Fig3  | √             | √   | 3days         | 14days  |
| JK30-500    | 14.0           | 23.7   | 3      | 10.2   | 20 AWG/ Φ0.8     | Fig3  | √             | √   | 3days         | 14days  |
| JK30-600    | 17.2           | 27     | 3      | 10.2   | 20 AWG/ Φ0.8     | Fig3  | √             | √   | 3days         | 14days  |
| JK30-700    | 17.2           | 27     | 3      | 10.2   | 20 AWG/ Φ0.8     | Fig3  | √             | √   | 3days         | 14days  |
| JK30-800    | 23.5           | 29.2   | 3      | 10.2   | 20 AWG/ Φ0.8     | Fig3  | √             | √   | 3days         | 14days  |
| JK30-900    | 23.5           | 29.2   | 3      | 10.2   | 20 AWG/ Φ0.8     | Fig3  | √             | √   | 3days         | 14days  |

Note: Dimensions A,B,Cis the maximum size,D Values is typical tolernce of ±0.5mm

### thermal Derating Chart-IH(A)

| Maximum ambient operating temperatures °C |       |      |       |       |      |      |      |      |      |      |
|-------------------------------------------|-------|------|-------|-------|------|------|------|------|------|------|
| Part Number                               | -40℃  | -20℃ | 0℃    | 25℃   | 40℃  | 50℃  | 60℃  | 70℃  | 80℃  | 85℃  |
| JK30-050                                  | 0.72  | 0.65 | 0.57  | 0.5A  | 0.45 | 0.41 | 0.38 | 0.34 | 0.30 | 0.25 |
| JK30-075                                  | 1.08  | 0.97 | 0.86  | 0.75A | 0.68 | 0.62 | 0.57 | 0.51 | 0.45 | 0.37 |
| JK30-090                                  | 1.30  | 1.17 | 1.03  | 0.9A  | 0.81 | 0.74 | 0.69 | 0.61 | 0.54 | 0.45 |
| JK30-110                                  | 1.59  | 1.43 | 1.26  | 1.1A  | 1.0  | 0.91 | 0.84 | 0.74 | 0.67 | 0.55 |
| JK30-120                                  | 1.74  | 1.56 | 1.38  | 1.2A  | 1.09 | 0.99 | 0.92 | 0.81 | 0.73 | 0.6  |
| JK30-135                                  | 1.95  | 1.75 | 1.55  | 1.35A | 1.22 | 1.12 | 1.03 | 0.91 | 0.82 | 0.67 |
| JK30-160                                  | 2.32  | 2.08 | 1.84  | 1.6A  | 1.45 | 1.32 | 1.23 | 1.08 | 0.97 | 0.8  |
| JK30-185                                  | 2.68  | 2.40 | 2.12  | 1.85A | 1.68 | 1.53 | 1.42 | 1.25 | 1.12 | 0.92 |
| JK30-200                                  | 2.9   | 2.6  | 2.3   | 2A    | 1.82 | 1.66 | 1.54 | 1.36 | 1.22 | 1    |
| JK30-250                                  | 3.62  | 3.25 | 2.87  | 2.5A  | 2.27 | 2.07 | 1.92 | 1.7  | 1.52 | 1.25 |
| JK30-300                                  | 4.35  | 3.9  | 3.45  | 3A    | 2.73 | 2.49 | 2.31 | 2.04 | 1.83 | 1.5  |
| JK30-400                                  | 5.8   | 5.2  | 4.6   | 4A    | 3.64 | 3.32 | 3.08 | 2.72 | 2.44 | 2    |
| JK30-500                                  | 7.25  | 6.5  | 5.75  | 5A    | 4.55 | 4.15 | 3.85 | 3.4  | 3.05 | 2.5  |
| JK30-600                                  | 8.7   | 7.8  | 6.9   | 6A    | 5.46 | 4.98 | 4.62 | 4.08 | 3.66 | 3    |
| JK30-700                                  | 10.15 | 9.1  | 8.05  | 7A    | 6.37 | 5.81 | 5.39 | 4.76 | 4.27 | 3.5  |
| JK30-800                                  | 11.6  | 10.4 | 9.2   | 8A    | 7.28 | 6.64 | 6.16 | 5.44 | 4.88 | 4    |
| JK30-900                                  | 13.05 | 11.7 | 10.35 | 9A    | 8.19 | 7.47 | 6.93 | 6.12 | 5.49 | 4.5  |

## Electrical characteristics(25°C)



| Part Number | I Hold | I Trip | V <sub>max</sub> | I <sub>max</sub> | P <sub>d</sub> Max | Maximum Time to Trip |          | Nominal resistance (mΩ) |                  | Certification |     | Delivery Time |         |
|-------------|--------|--------|------------------|------------------|--------------------|----------------------|----------|-------------------------|------------------|---------------|-----|---------------|---------|
|             | A      | A      | DC               | A                | W                  | Current (A)          | Time (S) | R <sub>min</sub>        | R <sub>max</sub> | UL            | TUV | in stock      | Produce |
| JK30-050    | 0.50   | 1.0    | 30V              | 40               | 0.5                | 2.5                  | 5.0      | 250                     | 600              | -             | √   | 3days         | 14days  |
| JK30-075    | 0.75   | 1.5    | 30V              | 40               | 0.6                | 3.75                 | 5.0      | 200                     | 370              | √             | √   | 3days         | 14days  |
| JK30-090    | 0.90   | 1.8    | 30V              | 40               | 0.7                | 4.5                  | 8.0      | 100                     | 220              | √             | √   | 3days         | 14days  |
| JK30-110    | 1.10   | 2.2    | 30V              | 40               | 0.7                | 5.5                  | 8.0      | 70                      | 200              | √             | √   | 3days         | 14days  |
| JK30-120    | 1.20   | 2.4    | 30V              | 40               | 0.8                | 6.0                  | 8.0      | 80                      | 180              | -             | √   | 3days         | 14days  |
| JK30-135    | 1.35   | 2.7    | 30V              | 40               | 0.8                | 6.75                 | 8.0      | 70                      | 160              | √             | √   | 3days         | 14days  |
| JK30-160    | 1.60   | 3.2    | 30V              | 40               | 0.9                | 8.0                  | 8.0      | 60                      | 140              | √             | √   | 3days         | 14days  |
| JK30-185    | 1.85   | 3.7    | 30V              | 40               | 1.0                | 9.25                 | 8.0      | 50                      | 120              | √             | √   | 3days         | 14days  |
| JK30-200    | 2.00   | 4.0    | 30V              | 40               | 1.2                | 10.0                 | 11       | 40                      | 100              | √             | √   | 3days         | 14days  |
| JK30-250    | 2.50   | 5.0    | 30V              | 40               | 1.2                | 12.5                 | 11       | 30                      | 80               | √             | √   | 3days         | 14days  |
| JK30-300    | 3.00   | 6.0    | 30V              | 40               | 2.0                | 15.0                 | 11       | 30                      | 70               | √             | √   | 3days         | 14days  |
| JK30-400    | 4.00   | 8.0    | 30V              | 40               | 2.5                | 20.0                 | 12.7     | 10                      | 60               | √             | √   | 3days         | 14days  |
| JK30-500    | 5.00   | 10     | 30V              | 40               | 3.0                | 25.0                 | 14.5     | 10                      | 50               | √             | √   | 3days         | 14days  |
| JK30-600    | 6.00   | 12     | 30V              | 40               | 3.5                | 30.0                 | 16       | 5                       | 40               | √             | √   | 3days         | 14days  |
| JK30-700    | 7.00   | 14     | 30V              | 40               | 3.8                | 35.0                 | 17.5     | 5                       | 30               | √             | √   | 3days         | 14days  |
| JK30-800    | 8.00   | 16     | 30V              | 40               | 4.0                | 40.0                 | 18.8     | 5                       | 25               | √             | √   | 3days         | 14days  |
| JK30-900    | 9.00   | 18     | 30V              | 40               | 4.2                | 40.0                 | 20       | 5                       | 20               | √             | √   | 3days         | 14days  |

I<sub>Hold</sub>=Hold current:maximum current at which the device will not trip at 25°C still air.

I<sub>Trip</sub>=Trip current:minimum current at which the device will nalways at 25°C still air.

V<sub>max</sub>=Maximum voltage device can withstand without damage at rated current.

I<sub>max</sub>=Maximum fault current device can withstand tithout damage at rated voltage.

T<sub>trip</sub>=Maximum time to trip(s) at assigned current.

P<sub>d</sub>=Typical power dissipation:typical amount of power dissipated by the decice when in state air environment.

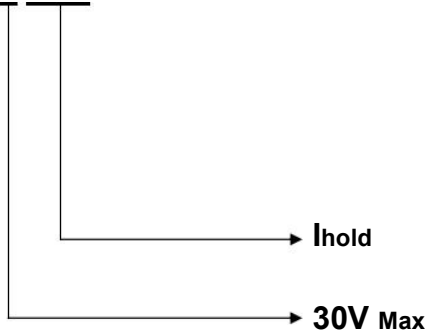
R<sub>min</sub>=Minimum device resistance at 25°C prior to tripping.

R<sub>max</sub>=Maximum device resistance at 25°C prior to tripping.

## Marking System



JK30 -XXX



## Test Procedures and Requirements

| Test            | Test Conditions                    | Accept/Reject Criteria        |
|-----------------|------------------------------------|-------------------------------|
| Resistance      | In still air @ 25℃                 | $R_{min} \leq R \leq R_{max}$ |
| Time to Trip    | Specified current, $V_{max}$ , 25℃ | Tmaximum Time to Trip         |
| Hold Current    | 60min ,at $I_H$                    | No trip                       |
| Trip Cycle Life | $V_{max}$ , $I_{max}$ , 100cycles  | No arcing or burning          |
| Trip Endurance  | $V_{max}$ , 24hours                | No arcing or burning          |

## Physical Characteristics and Environmental Specifications

## Physical Characteristics

| Test                  | Conditions              | Resistance change |
|-----------------------|-------------------------|-------------------|
| Passive aging         | +85℃, 1000hrs           | ±8% typical       |
| Humidity aging        | +85℃, 85%R.H.1000hrs    | ±8% typical       |
| Thermal shock         | +125℃ to -55℃, 10times  | ±12% typical      |
| Resistance to solvent | MIL-STD-202, Method 215 | No change         |
| Vibration             | MIL-STD-202, Method 201 | No change         |

## Operation Condition

1 Ambient temperature: -40℃~+85℃

2 Humidity: ≤95%HR(40℃)

3 Atmospheric pressure: 86Kpa ~106Kpa.

4 Vibration frequency: 10Hz ~ 50Hz.

5 Acceleration: 98m/s<sup>2</sup>.

6 Storage temperature: -40℃~85℃.

7Soldering

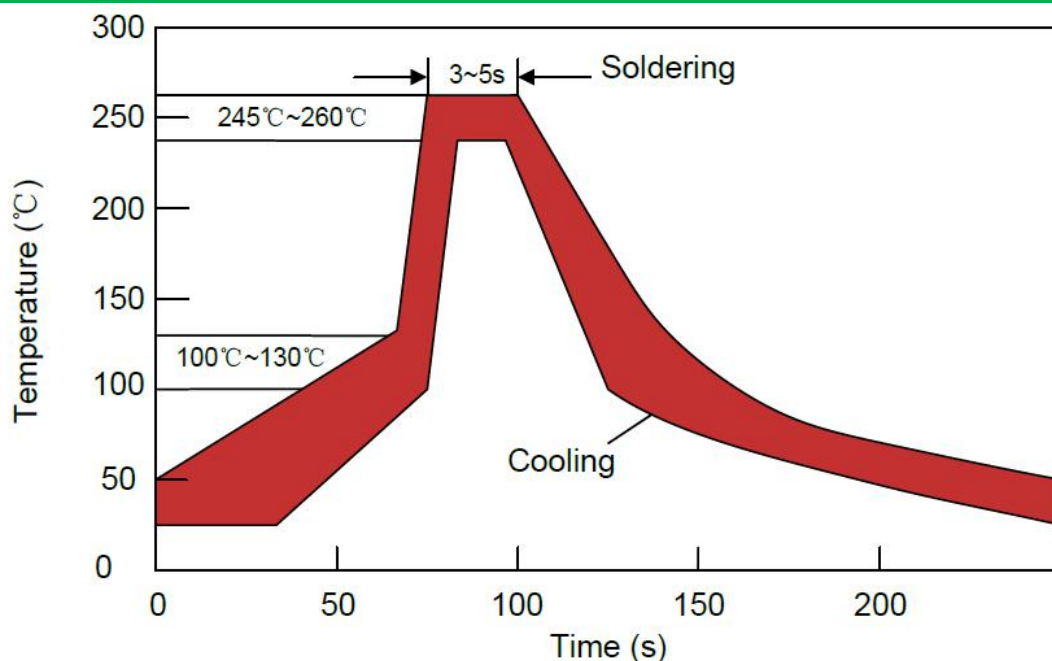
7.1 Wave Soldering:

Soldering Temperature:240℃~270℃

Soldering Time:≤5sec.

Soldering Position: Resettable fuse wire and the bottom ≥ 6mm。

## Wave Soldering Recommendation Parameters



### 7.2 Manual soldering

Soldering Temperature: 280°C~300°C

Soldering Time: ≤2sec.

Soldering Position: Resettable fuse wire and the bottom ≥ 6mm.

### Packing quantity:

JK30-050 ~ JK30-250 1000pcs/Bag

JK30-300 ~ JK30-500 500pcs/Bag

JK30-600 ~ JK30-900 200pcs/Bag



### Warning:

PTC devices are intended for protection against occasional over-current or over-temperature fault conditions, and should not be used when repeated fault conditions are anticipated. Operation beyond maximum ratings of improper use may result in device damage and possible electrical arcing and flame.

### Notes:

The specification is intended to present application, product and technical data to assist the user in selecting PTC circuit production devices. However, users should independently evaluate and test the suitability of each product. HUAAN makes no warranties as to the accuracy or completeness of the information and disclaims any liability resulting from its use. HUAAN's only obligations are those in the HUAAN Standard Terms and Conditions of Sale and in no case will HUAAN be liable for any incidental, indirect, or consequential damages arising from the sale, resale, or misuse of its products. HUAAN reserves the right to change or update, without notice, any information contained in this specification.